PYTHON FOR ACTUARIES · NOTEBOOK 07

Machine Learning for Actuaries

A 90-Minute Practical Workshop

In this session, you will learn how a model learns from training data and predicts new data. You will also meet the four models of the fraud detection exercise in Notebook 09.

- **Part 1**: Training and Predicting
- **Part 2**: Four Models, Same Steps
- **Part 3**: Training Data vs. Test Data
- **Part 4**: Outlook: When Accuracy Misleads

The explanations use the small tables `claims` and `new_claims`. All exercises use `vehicle_fraud.csv` from Notebook 03.

```
# Import required libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.neural_network import MLPClassifier
from sklearn.dummy import DummyClassifier
from xgboost import XGBClassifier
from sklearn.metrics import accuracy_score

# Avoid truncation of DataFrame output in Jupyter
pd.set_option('display.expand_frame_repr', False)

# Display all columns in the DataFrame output
pd.set_option('display.max_columns', None)

# Define a separator for better readability
sep = "=" * 50
```

Part 1: Training and Predicting

A model learns from past claims whose outcome is known: the **training data**. It then predicts the outcome of new claims: the **test data**. In the exercises, `train_test_split()` creates both sets (Notebook 05).

1.1 Past and New Claims

`claims` holds 16 past claims, `new_claims` four new ones. The features are already prepared as in Notebook 05. The outcome `fraud` of the new claims is known here, so the predictions can be checked.

Table	Role
`claims`	Training data: the model learns from these claims

Table	Role
`new_claims`	Test data: the model predicts these claims

```
# Past claims: the training data
claims = pd.DataFrame({
    'claim_value_ratio': [0.08, 0.12, 0.45, 0.10, 0.25, 0.06, 0.38, 0.15, 0.09, 0.52, 0.18,
        0.11, 0.07, 0.30, 0.14, 0.20],
    'reporting_delay_days': [5, 12, 38, 8, 15, 3, 30, 20, 6, 42, 10, 25, 4, 18, 9, 33],
    'police_report': [1, 1, 0, 1, 1, 1, 0, 0, 1, 0, 1, 1, 1, 0, 1, 1],
    'fraud': [0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 0, 0]
})

print(f"\n{sep}\nPast Claims: claims\n{sep}")
print(claims)
```

```
==================================================
Past Claims: claims
==================================================
    claim_value_ratio  reporting_delay_days  police_report  fraud
0                0.08                     5              1      0
1                0.12                    12              1      0
2                0.45                    38              0      1
3                0.10                     8              1      0
4                0.25                    15              1      0
5                0.06                     3              1      0
6                0.38                    30              0      1
7                0.15                    20              0      0
8                0.09                     6              1      0
9                0.52                    42              0      1
10               0.18                    10              1      0
11               0.11                    25              1      1
12               0.07                     4              1      0
13               0.30                    18              0      1
14               0.14                     9              1      0
15               0.20                    33              1      0
```

```
# New claims: the test data
new_claims = pd.DataFrame({
    'claim_value_ratio': [0.48, 0.10, 0.22, 0.16],
    'reporting_delay_days': [35, 7, 28, 14],
    'police_report': [0, 1, 1, 1],
    'fraud': [1, 0, 0, 1]
})

print(f"\n{sep}\nNew Claims: new_claims\n{sep}")
print(new_claims)
```

```
==================================================
New Claims: new_claims
==================================================
   claim_value_ratio  reporting_delay_days  police_report  fraud
0               0.48                    35              0      1
1               0.10                     7              1      0
2               0.22                    28              1      0
3               0.16                    14              1      1
```

1.2 Features, Target and Scaling

As in Notebook 05, `X` holds the features and `y` the target `fraud`. `X_new` and `y_new` hold the same for the new claims. The scaler learns from the past claims and scales both tables.

```
# Features and target of the past claims
X = claims.drop(columns=['fraud'])
y = claims['fraud']

print(f"\n{sep}\nFeatures: X.columns.tolist()\n{sep}")
print(X.columns.tolist())
```

```
================================================
Features: X.columns.tolist()
================================================
['claim_value_ratio', 'reporting_delay_days', 'police_report']
```

```
# Features and target of the new claims
X_new = new_claims.drop(columns=['fraud'])
y_new = new_claims['fraud']
```

```
# Learn from the past claims, apply to both tables
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_new_scaled = scaler.transform(X_new)
```

With real data, the other steps from Notebook 05 would come first: removing duplicates, handling impossible and missing values and encoding text columns. The small tables here need none of them, so this notebook skips them. The hands-on exercise in Notebook 09 goes through the full preparation.

1.3 Creating a Model

Logistic regression is a GLM for 0/1 targets: it turns a weighted sum of the features into a probability.

Creating a model sets its **hyperparameters** in parentheses: settings chosen before training. `max_iter=1000` allows enough steps for the computation, `random_state=42` makes the result reproducible. The new model has not learned anything yet.

```
# An untrained model with its hyperparameters
logreg = LogisticRegression(max_iter=1000, random_state=42)

print(f"\n{sep}\nModel: logreg\n{sep}")
print(logreg)
```

```
================================================
Model: logreg
================================================
LogisticRegression(max_iter=1000, random_state=42)
```

1.4 `fit()`: Learning from the Training Data

`fit()` learns from the features and the target of the training data. As with the `StandardScaler` in Notebook 05, the learned values are stored in the model, in attributes ending with `_`. Logistic regression learns one coefficient per feature: `coef_[0]`.

```
# Train the model on the past claims
logreg.fit(X_scaled, y)
print('Logistic Regression trained.')
```

```
Logistic Regression trained.
```

```
# Learned coefficients with the feature names
coefficients = pd.Series(logreg.coef_[0], index=X.columns)

print(f"\n{sep}\nCoefficients: logreg.coef_[0]\n{sep}")
print(coefficients.round(2))
```

```
==================================================
Coefficients: logreg.coef_[0]
==================================================
claim_value_ratio       0.68
reporting_delay_days    0.82
police_report          -0.62
dtype: float64
```

A high claim-to-value ratio and a late report raise the fraud probability, a police report lowers it.

1.5 `predict()`: Predicting New Claims

`predict()` applies the learned coefficients to new claims and returns `1` (fraud) or `0` (no fraud) per claim. It needs only the features. New claims need the same preparation as the training data: here, the scaled features `X_new_scaled`.

```
# Predict the new claims
y_pred = logreg.predict(X_new_scaled)

print(f"\n{sep}\nPredictions: logreg.predict(X_new_scaled)\n{sep}")
print(y_pred)
```

```
==================================================
Predictions: logreg.predict(X_new_scaled)
==================================================
[1 0 0 0]
```

1.6 `predict_proba()`: Predicting Probabilities

`predict_proba()` returns two columns per claim: the probability of `0` and the probability of `1`. Each row sums to 1.

```
# Probabilities of 0 and 1 per claim
print(f"\n{sep}\nProbabilities: logreg.predict_proba(X_new_scaled)\n{sep}")
print(logreg.predict_proba(X_new_scaled).round(3))
```

```
================================================
Probabilities: logreg.predict_proba(X_new_scaled)
================================================
[[0.087 0.913]
 [0.938 0.062]
 [0.676 0.324]
 [0.877 0.123]]
```

`[:, 1]` selects the second column: the fraud probability. `predict()` turns it into `0` or `1` with the threshold 0.5.

```
# Fraud probability per claim
y_proba = logreg.predict_proba(X_new_scaled)[:, 1]
```

```
# Fraud probability, prediction and actual outcome
comparison = pd.DataFrame({'fraud_probability': y_proba, 'prediction': y_pred, 'fraud':
    y_new})

print(f"\n{sep}\nComparison: fraud_probability, prediction, fraud\n{sep}")
print(comparison.round(3))
```

```
================================================
Comparison: fraud_probability, prediction, fraud
================================================
   fraud_probability  prediction  fraud
0              0.913           1      1
1              0.062           0      0
2              0.324           0      0
3              0.123           0      1
```

Claim 0 is flagged correctly. Claim 3 is fraud, but the model gives it only 12%: it looks like a legitimate claim.

Exercise 1: Training and Predicting on `vehicle_fraud.csv`

Train a logistic regression, read its coefficients and predict classes and fraud probabilities for the test data.

The first four code cells prepare the data in a shortened but clean way: they remove the duplicates, use only six numeric features without missing values and scale after the train/test split. Notebook 09 shows the full preparation from Notebook 05.

```
# Load the dataset and remove the duplicates
df = pd.read_csv('vehicle_fraud.csv')
df = df.drop_duplicates()
```

```
# Six features without missing values
df['claim_value_ratio'] = df['claim_amount'] / df['vehicle_value']
features = ['claim_value_ratio', 'reporting_delay_days', 'customer_since_years',
            'number_of_prior_claims', 'police_report', 'witnesses_present']
```

```
# 80% training data, 20% test data
X_train, X_test, y_train, y_test = train_test_split(df[features], df['fraud'],
    test_size=0.2, random_state=42, stratify=df['fraud'])
```

```
# Learn from the training data, apply to both sets
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Task 1: Create a logistic regression `lr_model` with `max_iter=1000` and `random_state=42`. Train it on the scaled training data.

```
# Task 1: Logistic regression
# Hint: lr_model = LogisticRegression(max_iter=1000, random_state=42)
#       lr_model.fit(X_train_scaled, y_train)
```

Task 2: Show the learned coefficients with the feature names, sorted. Which feature raises the fraud probability most? Which features lower it?

```
# Task 2: Learned coefficients
# Hint: coefficients = pd.Series(lr_model.coef_[0], index=X_train.columns)
#       coefficients.sort_values()
```

Task 3: Predict the test data. How many claims does the model flag as fraud? How many test claims are fraudulent?

```
# Task 3: Predictions
# Hint: y_pred = lr_model.predict(X_test_scaled)
#       y_pred.sum(), y_test.sum()
```

Task 4: Compute the fraud probabilities of the test data. Show the five claims with the highest probability together with `fraud`.

```
# Task 4: Fraud probabilities
# Hint: y_proba = lr_model.predict_proba(X_test_scaled)[:, 1]
#       comparison = pd.DataFrame({'fraud_probability': y_proba, 'fraud': y_test})
#       comparison.sort_values('fraud_probability', ascending=False).head()
```

Task 5: A new claim: claim amount 1.5 times the vehicle value, reported after 35 days, customer for 1 year, 3 prior claims, no police report, no witnesses. Predict its fraud probability.

```
# Task 5: Fraud probability of a new claim
# Hint: new_claim = pd.DataFrame({'claim_value_ratio': [1.5], 'reporting_delay_days': [35],
#                                 'customer_since_years': [1], 'number_of_prior_claims':
    [3],
#                                 'police_report': [0], 'witnesses_present': [0]})
#       new_claim_scaled = scaler.transform(new_claim)
#       lr_model.predict_proba(new_claim_scaled)[:, 1]
```

Part 2: Four Models, Same Steps

Every model is used the same way: create it, train it with `fit()`, predict with `predict()` and `predict_proba()`. Only the line that creates the model changes. Notebook 09 uses these four models:

Model	Idea	Features
`LogisticRegression`	GLM for 0/1 targets	scaled
`RandomForestClassifier`	Many decision trees that vote	unscaled
`XGBClassifier`	Decision trees that correct each other	unscaled
`MLPClassifier`	Neural network	scaled

As in Notebook 05: logistic regression and neural networks need scaled features, tree-based models do not.

2.1 Random Forest

A decision tree sorts the claims with yes/no questions, e.g. "reported after more than 20 days?". A random forest trains many trees, each on a random sample of the claims, and averages their votes. `n_estimators=100` sets the number of trees.

```
# Random forest with 100 trees
forest = RandomForestClassifier(n_estimators=100, random_state=42)
forest.fit(X, y)
print('Random Forest trained.')
```

```
Random Forest trained.
```

2.2 XGBoost

XGBoost also uses decision trees but builds them one after another: each new tree corrects the errors of the trees before it (**gradient boosting**). `n_estimators=100` sets the number of trees.

```
# XGBoost with 100 trees
boosting = XGBClassifier(n_estimators=100, random_state=42)
boosting.fit(X, y)
print('XGBoost trained.')
```

```
XGBoost trained.
```

2.3 Neural Network

A neural network passes the features through layers of neurons. `hidden_layer_sizes=(64, 32)` sets two layers with 64 and 32 neurons, `max_iter=500` limits the number of training rounds. `early_stopping=True` holds back 10% of the training data and stops training when the predictions there no longer improve.

```
# Neural network with two layers
network = MLPClassifier(hidden_layer_sizes=(64, 32), max_iter=500, early_stopping=True,
    random_state=42)
network.fit(X_scaled, y)
print('Neural Network trained.')
```

```
Neural Network trained.
```

2.4 Comparing the Models

The dictionary `models` stores each model with the features it needs. The loop `for name, (model, X_eval) in models.items()` takes one model with its features at a time.

```
# Models with the features of the new claims
models = {
    'Logistic Regression': (logreg, X_new_scaled),
    'Random Forest': (forest, X_new),
    'XGBoost': (boosting, X_new),
    'Neural Network': (network, X_new_scaled),
}
```

```
# Fraud probabilities of the new claims from every model
probas = {}
for name, (model, X_eval) in models.items():
    probas[name] = model.predict_proba(X_eval)[:, 1]
probas['fraud'] = y_new
probas_df = pd.DataFrame(probas)

print(f"\n{sep}\nFraud Probabilities of the New Claims\n{sep}")
print(probas_df.round(3))
```

```
================================================
Fraud Probabilities of the New Claims
================================================
   Logistic Regression  Random Forest  XGBoost  Neural Network  fraud
0                0.913           0.94    0.803           0.506      1
1                0.062           0.03    0.067           0.278      0
2                0.324           0.43    0.493           0.412      0
3                0.123           0.00    0.067           0.334      1
```

All four models flag claim 0 and miss claim 3, but they differ in how certain they are. The neural network is the least certain: 16 claims are far too few for it.

Exercise 2: Four Models on `vehicle_fraud.csv`

Train a random forest, XGBoost and a neural network with the same steps and compare their fraud probabilities.

Continue with the data from Exercise 1. Use `random_state=42` for every model.

Task 1: Train a random forest `rf_model` with `n_estimators=100`.

```
# Task 1: Random forest
# Hint: rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
#       rf_model.fit(X_train, y_train)
```

Task 2: Train XGBoost `xgb_model` with `n_estimators=100`.

```
# Task 2: XGBoost
# Hint: xgb_model = XGBClassifier(n_estimators=100, random_state=42)
#       xgb_model.fit(X_train, y_train)
```

Task 3: Train a neural network `mlp_model` with `hidden_layer_sizes=(64, 32)`, `max_iter=500` and `early_stopping=True`.

```
# Task 3: Neural network
# Hint: mlp_model = MLPClassifier(hidden_layer_sizes=(64, 32), max_iter=500,
    early_stopping=True, random_state=42)
#       mlp_model.fit(X_train_scaled, y_train)
```

Task 4: Store the four models with their test data in the dictionary `models`. Show the fraud probabilities of all four models for the first ten test claims together with `fraud`. Then filter the fraud cases that all four models miss: `fraud` is `1`, but the highest of the four probabilities is below 0.5. How many are there?

```
# Task 4: Fraud probabilities of all four models
# Hint: models = {
#           'Logistic Regression': (lr_model, X_test_scaled),
#           'Random Forest': (rf_model, X_test),
#           'XGBoost': (xgb_model, X_test),
#           'Neural Network': (mlp_model, X_test_scaled),
#       }
#       probas = {}
#       for name, (model, X_eval) in models.items():
#           probas[name] = model.predict_proba(X_eval)[:, 1]
#       probas['fraud'] = y_test
#       probas_df = pd.DataFrame(probas)
#       probas_df.head(10)
# And later, to filter the fraud cases that all four models miss:
#       highest = probas_df.drop(columns='fraud').max(axis=1)
#       missed = probas_df[(probas_df['fraud'] == 1) & (highest < 0.5)]
#       len(missed), missed.head()
```

Part 3: Training Data vs. Test Data

3.1 Accuracy

`accuracy_score(y_true, y_pred)` compares the predictions with the actual outcomes and returns the share of correct predictions.

```
# Share of correct predictions for the new claims
print(f"\n{sep}\nAccuracy: accuracy_score(y_new, logreg.predict(X_new_scaled))\n{sep}")
print(accuracy_score(y_new, logreg.predict(X_new_scaled)))
```

```
================================================
Accuracy: accuracy_score(y_new, logreg.predict(X_new_scaled))
================================================
0.75
```

The logistic regression predicts three of the four new claims correctly.

3.2 Learning by Heart

A model can learn its training data by heart (**overfitting**). Its accuracy on the training data then says little about new claims. The random forest predicts the past claims it was trained on, then the new claims.

```
# Accuracy on the past claims
print(f"\n{sep}\nAccuracy: accuracy_score(y, forest.predict(X))\n{sep}")
print(accuracy_score(y, forest.predict(X)))
```

```
================================================
Accuracy: accuracy_score(y, forest.predict(X))
================================================
1.0
```

```
# Accuracy on the new claims
print(f"\n{sep}\nAccuracy: accuracy_score(y_new, forest.predict(X_new))\n{sep}")
print(accuracy_score(y_new, forest.predict(X_new)))
```

```
================================================
Accuracy: accuracy_score(y_new, forest.predict(X_new))
================================================
0.75
```

On the past claims, the forest is perfect: it even recognizes claim 11, a fraud case that looks legitimate. On the new claims, it reaches 75%. Only data the model has not seen during training show how well it predicts. That is why every model is evaluated on the test data.

> **Exercise 3: Training Data vs. Test Data on `vehicle_fraud.csv`**

Measure the accuracy on test and training data and compare it with a model that always predicts "no fraud".

Continue with `models` from Exercise 2.

Task 1: Compute the accuracy of every model on the test data.

```
# Task 1: Accuracy on the test data
# Hint: for name, (model, X_eval) in models.items():
#           print(name, accuracy_score(y_test, model.predict(X_eval)))
```

Task 2: Compute the accuracy of the random forest on the training data. Compare it with Task 1.

```
# Task 2: Accuracy of the random forest on the training data
# Hint: accuracy_score(y_train, rf_model.predict(X_train))
```

Task 3: A model that always predicts "no fraud" is right for every legitimate claim. What accuracy does it reach on the test data? Compare it with Task 1.

```
# Task 3: Accuracy of a model that always predicts "no fraud"
# Hint: (y_test == 0).mean()
```

Part 4: Outlook: When Accuracy Misleads

Fraud is rare: only about 12% of the claims in `vehicle_fraud.csv` are fraudulent. With such **imbalanced** data, a high accuracy can come from a model that never finds a single fraud case. This part is an outlook; the seminar does not go deeper into it.

4.1 A Baseline That Always Predicts "No Fraud"

`DummyClassifier(strategy='most_frequent')` ignores the features and always predicts the most frequent class of the training data, here `0` (no fraud). A real model has to beat this baseline.

```
# A baseline that always predicts the most frequent class
dummy = DummyClassifier(strategy='most_frequent')
dummy.fit(X, y)

print(f"\n{sep}\nPredictions: dummy.predict(X_new)\n{sep}")
print(dummy.predict(X_new))
```

```
================================================
Predictions: dummy.predict(X_new)
================================================
[0 0 0 0]
```

```
# Accuracy of the baseline on the past and the new claims
print(f"\n{sep}\nAccuracy on the Past Claims: accuracy_score(y, dummy.predict(X))\n{sep}")
print(accuracy_score(y, dummy.predict(X)))

print(f"\n{sep}\nAccuracy on the New Claims: accuracy_score(y_new,
    dummy.predict(X_new))\n{sep}")
print(accuracy_score(y_new, dummy.predict(X_new)))
```

```
================================================
Accuracy on the Past Claims: accuracy_score(y, dummy.predict(X))
================================================
0.6875

================================================
Accuracy on the New Claims: accuracy_score(y_new, dummy.predict(X_new))
================================================
0.5
```

The baseline is right for 11 of the 16 past claims (69%) without catching a single fraud case. On the new claims, it reaches 50% compared with 75% for the logistic regression. In `vehicle_fraud.csv`, where fraud is rarer, the same baseline even reaches 88% (Exercise 3, Task 3).

4.2 Precision, Recall and F1-Score

Three measures focus on the fraud cases instead:

Measure	Question	Dummy baseline
Precision	Of the claims flagged as fraud, how many are fraud?	not defined: it flags no claim
Recall	Of all fraud cases, how many are flagged?	0
F1-score	Balance of precision and recall	0

Notebook 09 compares the models with these measures: `precision_score()`, `recall_score()` and `f1_score()`.

Summary

- **Training and predicting:** create a model with its hyperparameters, `fit()` on the training data, `predict()` and `predict_proba()` on new data with the same preparation
- **Four models:** `LogisticRegression`, `RandomForestClassifier`, `XGBClassifier` and `MLPClassifier`, all used the same way
- **Training vs. test data:** `accuracy_score()`, evaluation on the test data only
- **Outlook:** on imbalanced data, a baseline that always predicts "no fraud" reaches a high accuracy; precision, recall and F1-score focus on the fraud cases (Notebook 09)

Next: your choice between GLMs for the average claim amount in Notebook 08 and a session on LLMs, then detecting fraudulent claims in Notebook 09.

- scikit-learn: Getting Started
- XGBoost Documentation